

# Punjaya Wickramasinghe

## Mechatronics Engineer


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## WORK EXPERIENCE

**Software Engineering Lead**
Phoenix ICT Solutions | 2025–2026

LangChain | Ollama | FastAPI | Docker | VirtualBox | Flutter

- Co-led a team of 5 to architect and implement an air gapped multi-purpose AI agent-enabled solution (question-answering, analysis, report generation, intelligent referencing) to accelerate user workflows
- Deployed comprehensive evaluation pipelines to achieve the following scores: correctness 90%, groundedness 97%, context recall 78%, source relevance 83%
- Involved with deployment into air gapped environment

**Engineering Fellow**
Engineering for Change & Resham Sutra | 2025

Java | Figma | Python | C++ | Whisper.cpp

- Designed and integrated AI voice assistant into India's top textile marketplace app to improve UX for low-literacy users
- Packaged into a 40 MB offline module, the solution runs on low-end Android devices
- Applying human-centered design to create intuitive UX to overcome language and literacy barriers
- Ensuring scalability of AI voice solutions to regional languages (Hindi, Marathi) to boost reach and engagement

**Mechatronics Engineer**
Geodrones Australia | 2024

C++ | ROS | Gazebo | Docker | Altium | STM32 | PyTorch

- Finalist, Australian AI Awards 2024
- Deployed real-time drone-based AI system integrating camera/LiDAR for live change detection (C++, PyTorch)
- Developed secure real-time task swapping, significantly enhancing drone operational flexibility (Docker)
- Adapted open-source GPS-denied UAV state estimator for proprietary drones (ROS/Gazebo)
- Designed PCBs & battery management system (Altium, STM32)
- Pitched demo to industry VIPs at October Sky 2024

## EDUCATION

**Bachelor of Engineering (Research & Development)**
ANU | 2019 – 2023

- Mechatronics & Comp Science
- GPA: 6.72 / 7
- First Class Honours
- Excellence Scholarship (2019)
- CECC Dickins Scholarship (2023)

**ATAR: 99.20 (≈ IB Score: 44/45)**
Narrabundah College | 2018 – 2019

## EXTRA CURRICULAR

**Head of Social**
ANU Engineers Assoc. | 2020 – 2021

- ANU's 3rd largest student body
- Led a team of 3 to run 5 social events, generating ~$2,000 in sales, exceeding previous year's sales

**Community Volunteer**
Aus Sri Lankan Buddhist Assoc. | 2010–

- MC at cultural events
- Regular volunteer to assist with community events (logistics, photographer)

**National Premier League Football**
Player, 2013–

- Played highest level in Canberra until 2020
- Runners-up in 2019 (U18 NPL1) and semi-finalists in 2021 (U23 NPL2)
- Consistently trained while managing a balanced lifestyle

## RESEARCH EXPERIENCE

### Novel INS System for Improved UAV State Estimation
ANU R&D Student | 2023

Lie Theory | Python | LaTeX

- Co-authored paper in European Journal of Control – "Constructive synchronous observer design for inertial navigation with delayed GNSS measurements".
- Achieved 86/100 on thesis and seminar.
- Advanced linear algebra to execute the implementation of a novel INS filter (ArduPilot, Python).
- Fused sensor measurements (magnetometer, barometer, GPS velocity) to further enhance performance.
- Results of my initial implementation were presented at the ArduPilot 2023 Conference.

### Acoustic Emotion Recognition with Machine Learning
ANU R&D Student | 2021

sklearn | TensorFlow | Seaborn | GridSearchCV | librosa

- Investigated spatial audio features for enhancing acoustic emotion recognition with ML (Python, MATLAB).
- Achieved 91/100 on thesis and seminar.
- Applied knowledge in spatial audio and HRTF concepts (librosa)
- Data visualisation and ML algorithm adaptation (Seaborn)
- Implemented KNN, MLP and CNN (sklearn, TensorFlow)

### Metal-Organic Frameworks for Acetone Breath Sensors
ANU R&D Student | 2020

- Achieved 88/100 on thesis and seminar
- Meta-analysis to find areas to improve acetone breath sensors

## OTHER EXPERIENCE

### BixeLab
Testing Analyst | 2021 – 2023

- Data analysis on biometric tests to curate client reports
- Automated processes and handled risk management
- World's second NIST/FIDO accredited biometric testing lab
- Small team of 15 - demanded adaptability and learning agility

### Private Tutor
Math & Physics | 2019–

- Tutored 55 students since 2019, Yr 3 to 12, with diverse backgrounds
- Improved student grades from C/B to A average
- Simplified complex concepts for easy understanding
- Taught time management and self-sufficiency skills
- Focus on instilling lasting subject curiosity in students




## REFERENCES

**Darcy Slevin**
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Provolution Health / CEO
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